^{36}Ar(d,α),(pol d,α) 1969Br21,1972Ho02,1982Ta04

^{36}Ar→1p+1n+^{34}Cl from $J^\pi=0^+$ ^{36}Ar ground state.

1972Ho02: a 19-MeV deuteron beam bombarded a 99.6%-enriched ^{36}Ar gas target. Outgoing α particles were momentum-analyzed by the Enge split-pole spectrograph and detected in 100−μm Ilford K0 nuclear emulsion plates. Measured $\sigma(E_\alpha,\theta)$. Deduced levels, L-transfers, L-mixings, J, π, and isospin with DWUCK-DWBA analysis and χ^2 versus L+L′ analysis.

1982Ta04,1980To13: a 16-MeV polarized deuteron beam from the TUNL Lamb-shift ion source bombarded a 99.9%-enriched ^{36}Ar target. Outgoing α particles were detected in four solid-state detectors with FWHM=100-200 keV. Measured $\sigma(E_\alpha,\theta)$, vector analyzing powers, $iT_{11}(E_\alpha,\theta)$, and tensor analyzing powers, $T_{20}(E_\alpha,\theta)$ and $T_{22}(E_\alpha,\theta)$. Deduced levels, L-transfers, J, π, and isospin with DWUCK-DWBA analysis.

1969Br21: a 45-MeV deuteron beam was produced from the Berkeley 88-Inch Cyclotron. The target was 99.6%-enriched ^{36}Ar gas. Reaction products were detected using a 155−μm phosphorus-diffused silicon ΔE transmission counter and a 3.0-mm lithium-drifted silicon E counter operated in coincidence, followed by a 0.5-mm lithium-drifted silicon E-reject counter operated in anti-coincidence with the first two counters to eliminate long-range particles. Measured $\sigma(E_\alpha,\theta)$. Deduced levels, L-transfers, J, π, and isospin. Also measured ^{36}Ar(p,t)^{34}Ar and ^{36}Ar(p,^{3}He)^{34}Cl.

Others: Measured cross section of ^{36}Ar(d,α)^{34m}Cl, E≤8.4 MeV (2011En01,2012En01).

^{34}Cl Levels

E(level)†	J^π‡	L#	Comments
0?			Very weakly at all angles due to parity selection rule and isospin selection rule of ^{36}Ar(d,α)^{34}Cl.
147 *20*	3^+	4	J^π: L-1 transfer from measured analyzing powers in 1982Ta04. L: 4 from 1972Ho02 and 1982Ta04.
459 *20*	1^+	2	J^π: L-1 transfer from measured analyzing powers in 1982Ta04. L: from 1982Ta04; other: L=0+2 in 1972Ho02.
665 *20*	$1^+,3^+$	0+2,2+4	
1229 *20*	2^+	2	J^π: L transfer from measured analyzing powers in 1982Ta04. L: from 1982Ta04; other: L=0+2 in 1972Ho02.
1890 *20*	2^+	2	J^π: L transfer from measured analyzing powers in 1982Ta04. L: from 1982Ta04; other: L=2+4,3+5 for an unresolved 1890 and 1920 doublet in 1972Ho02.
2191 *20*	3^+	(2)	J^π: L+1 transfer from measured analyzing powers in 1982Ta04. L: from 1982Ta04; other: L=2+4 or pure L=4 in 1972Ho02.
2382 *20*	1^+	0+2	
2589 *20*	1^+	0+2	
2620 *20*			
2725 *20*	2^-	1+3	
3146 *20*	1^+	0+2	
3334 *20*	$2^-,3^-,4^-$	3	E(level): other: 3330 *40* (1969Br21).
3377 *20*			
4970 *40*	$(0)^+$	0	E(level): doublet of T=0 and 1. E(level) is likely determined from ^{36}Ar(p,^{3}He)^{34}Cl data (1969Br21). T=1 component corresponds to 0^+.
5600 *40*			E(level): E(level) is likely determined from both ^{36}Ar(d,α)^{34}Cl and ^{36}Ar(p,^{3}He)^{34}Cl data (1969Br21).
6160 *40*	$(2)^+$	2	E(level): doublet of T=0 and 1. E(level) is likely determined from ^{36}Ar(p,^{3}He)^{34}Cl data (1969Br21). T=1 component corresponds to 2^+.
7070 *40*			E(level): E(level) is likely determined from ^{36}Ar(p,^{3}He)^{34}Cl data (1969Br21).

† From 1972Ho02, unless otherwise noted.
‡ Deduced from measured L values, unless otherwise noted.
# From DWUCK-DWBA analysis of the measured $\sigma(\theta)$ and χ^2 versus L+L′ analysis in 1972Ho02, unless otherwise noted.